

Lipid extractions are typically done by the Bligh and Dyer (B&D) method, which uses a water/methanol mixture and chloroform for the two phases of the solvent partition. Researchers specializing in green chemistry (GC) have proposed that greener solvents such as ethyl acetate and ethanol can be used to replace the toxic chloroform and methanol, respectively, from the B&D method. The following experiments test this proposal by comparing the B&D and GC solvent methods (Figure 1) to separate lipids from an extract of *Yarrowia lipolytica* yeast cells.

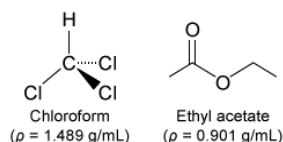


Figure 1 Structures and densities (ρ) of the B&D and GC organic solvents

Experiment 1

Y. lipolytica grown on rice straw containing 2,6-dimethoxyphenol were dried, then mixed with distilled water and 0.85% potassium chloride to create yeast extracts. Homogenized yeast extracts were transferred to separation funnels, and solutions with a 2:3.8 ratio of chloroform to methanol/water or ethyl acetate to ethanol/water were added. The resulting layers (L1, L2) in each extraction method were separated at the interface by films of cellular debris, as shown in Figure 2.

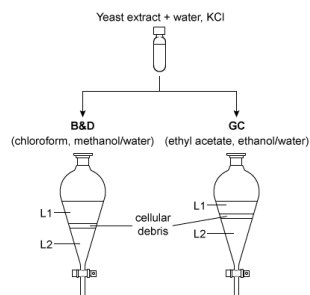


Figure 2 Separations obtained from B&D and GC lipid extraction methods

Experiment 2

Computational models indicate that glycerol (Figure 3) found in the yeast extract demonstrates partition differences between solvents in the B&D and GC methods.

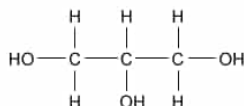


Figure 3 Structure of glycerol

To verify this prediction, chemists analyzed the B&D and GC solvent mixtures by proton nuclear magnetic resonance (^1H NMR) spectroscopy. A 160 mg/mL glycerol standard was separately prepared in each organic solvent from the B&D and GC methods. The glycerol standard was extracted with an equivalent volume of water/alcohol, then the organic layers were used to prepare NMR samples through iterative dilution. A 0.5 mL sample of organic layer was diluted with enough solvent to give a total volume of 10.0 mL, then 0.5 mL of this resulting solution was diluted with 0.5 mL of solvent to yield a final 1.0 mL NMR sample. The ^1H NMR spectra obtained from the two samples are shown in Figure 4 (hydroxyl hydrogens are not observed in the spectra).

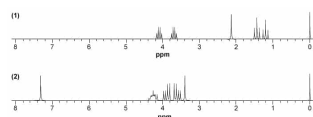


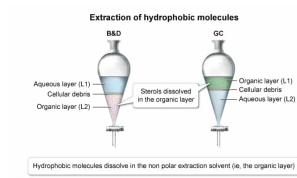
Figure 4 ^1H NMR spectra of the diluted organic solvent mixtures used in the (1) GC and (2) B&D extractions. The B&D mixture demonstrates the presence of glycerol.

Which layer in the B&D and GC methods in Figure 2 is expected to contain sterols isolated from the *Y. lipolytica* extract? (Note: Assume that the majority of the alcohols are retained in the aqueous layers.)

- ☐ A. L1 in the B&D method and L2 in the GC method [22%]
- ☒ B. L2 in the B&D method and L1 in the GC method [31%]
- ☐ C. L1 in the GC method and L1 in the B&D method [26%]
- ☐ D. L2 in the GC method and L2 in the B&D method [19%]

Correct
31% answered correctly

Explanation:



Extraction is a technique that uses **immiscible solvents** (ie, an organic solvent and an aqueous solution that do not dissolve in each other) as a partition to separate molecules in a mixture according to their **solubility**. Molecules dissolve best in solvents with a **similar polarity** (ie, "like dissolves like"). **Hydrophobic molecules** with few polar bonds have a **greater affinity** for the nonpolar **organic layer** whereas hydrophilic molecules with many polar bonds have a greater affinity for the polar aqueous phase.

In the solvents used for each extraction in Figure 2, the alcohols (methanol and ethanol) are miscible with water and the majority remain dissolved in the aqueous layer. Chloroform and ethyl acetate are significantly less polar and are immiscible in water; consequently, chloroform and ethyl acetate comprise the organic layers in the B&D and GC extractions, respectively. Sterols isolated from *Y. lipolytica* are molecules composed mostly of carbon and hydrogen. As such, sterols are predominantly hydrophobic and will migrate to the organic layer in both extractions.

However, the organic layers are in **different positions** in each extraction method according to the solvent density (ie, the densest solvent is always the lowest layer). Chloroform ($\rho = 1.5 \text{ g/mL}$) is denser than the aqueous layer ($\rho = 1.0 \text{ g/mL}$) whereas ethyl acetate ($\rho = 0.9 \text{ g/mL}$) is less dense than the aqueous layer. Therefore, the sterols isolated from *Y. lipolytica*, are found in L2 in the B&D method and L1 in the GC method.

(Choices A, C, and D) L1 and L2 are the aqueous layers of the B&D and GC methods, respectively.

Educational objective:

Extraction is a technique that uses an immiscible organic solvent and an aqueous solution as a partition to separate molecules in a mixture based on solubility. Molecules dissolve best in solvents with a similar polarity (ie, "like dissolves like"). Hydrophobic molecules have a greater affinity for the organic layer because the organic layer is less polar than the aqueous layer.

Time spent: 0 sec
Subject:
Organic Chemistry

QID: 401238
System:
Separation Techniques, Spectroscopy, and Analytical Methods

Lipid extractions are typically done by the Bligh and Dyer (B&D) method, which uses a water/methanol mixture and chloroform for the two phases of the solvent partition. Researchers specializing in green chemistry (GC) have proposed that greener solvents such as ethyl acetate and ethanol can be used to replace the toxic chloroform and methanol, respectively, from the B&D method. The following experiments test this proposal by comparing the B&D and GC solvent methods (Figure 1) to separate lipids from an extract of *Yarrowia lipolytica* yeast cells.

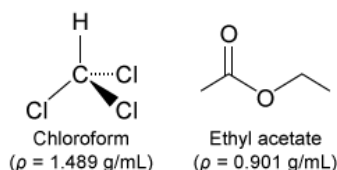


Figure 1 Structures and densities (ρ) of the B&D and GC organic solvents

Experiment 1

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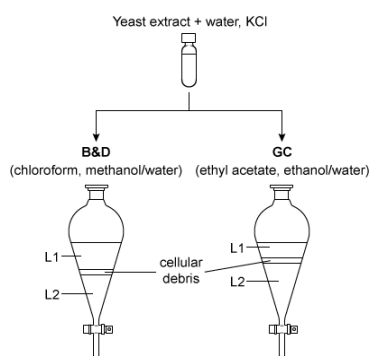


Figure 2 Separations obtained from B&D and GC lipid extraction methods

Experiment 2

Computational models indicate that glycerol (Figure 3) found in the yeast extract demonstrates partition differences between solvents in the B&D and GC methods.

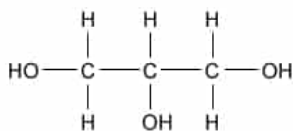


Figure 3 Structure of glycerol

To verify this prediction, chemists analyzed the B&D and GC solvent mixtures by proton nuclear magnetic resonance (^1H NMR) spectroscopy. A 160 mg/mL glycerol standard was separately prepared in each organic solvent from the B&D and GC methods. The glycerol standard was extracted with an equivalent volume of water/alcohol, then the organic layers were used to prepare NMR samples through iterative dilution. A 0.5 mL sample of organic layer was diluted with enough solvent to give a total volume of 10.0 mL, then 0.5 mL of this resulting solution was diluted with 0.5 mL of solvent to yield a final 1.0 mL NMR sample. The ^1H NMR spectra obtained from the two samples are shown in Figure 4 (hydroxyl hydrogens are not observed in the spectra).

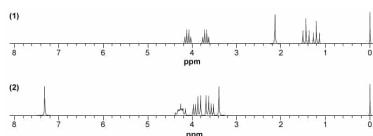


Figure 4 ^1H NMR spectra of the diluted organic solvent mixtures used in the (1) GC and (2) B&D extractions. The B&D mixture demonstrates the presence of glycerol.

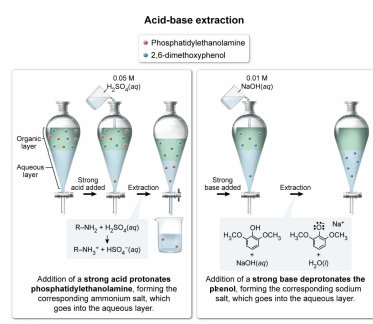
In Experiment 1, which aqueous solutions could be included in the extraction protocol to cause phosphatidylethanolamine, then 2,6-dimethoxyphenol to sequentially enter the aqueous layer?

- ☐ A. 0.05 M NaOH(aq), then 0.01 M H₂SO₄(aq). [20%]
- ☐ B. 0.05 M HCl(aq), then 0.01 M NaHCO₃(aq). [23%]
- ☐ C. 0.01 M H₂CO₃(aq), then 0.05 M NaHCO₃(aq). [16%]
- ✓ ☒ D. 0.05 M H₂SO₄(aq), then 0.01 M NaOH(aq). [38%]

Correct

38% answered correctly

Explanation:



A mixture of organic compounds with **acidic and basic functional groups** that are initially in the organic layer can be induced to enter the aqueous layer by converting them into **ionic salts**. Ionic salts are formed by either **deprotonation or protonation** of the compounds through treatment with an aqueous base or acid, respectively.

Strong bases form ionic salts easily when **weak acids** are added to a mixture. Conversely, **weaker bases** such as amines are only **protonated** by **strong acids**. Carboxylic acids can be deprotonated by both **strong and weak bases**; however, phenols are much weaker acids and require strong bases to be deprotonated.

Because the question states that phosphatidylethanolamine should be extracted to the aqueous layer first, followed by 2,6-dimethoxyphenol, their ionic salts must be formed in sequence. Phosphatidylethanolamine contains an amine group that can be protonated to form a water-soluble ammonium salt; therefore, a **strong acid** such as H₂SO₄ should be first used to **protonate the amino group**. In contrast, 2,6-dimethoxyphenol has a **weakly acidic** phenol group ($pK_a \approx 10$) that must be deprotonated to form an ionic salt. Only **strong bases** such as NaOH can **deprotonate phenols**.

Therefore, for phosphatidylethanolamine to be extracted into the aqueous layer before 2,6-dimethoxyphenol, the sequence of aqueous extractions must be addition of a strong acid (0.05 M H₂SO₄(aq)) followed by a strong base (0.01 M NaOH(aq)).

(Choice A) The addition of a strong base (NaOH) first will cause 2,6-dimethoxyphenol to be deprotonated and extracted *before* phosphatidylethanolamine.

(Choice B) Although the first step would successfully extract phosphatidylethanolamine, NaHCO₃(aq) is a weak base that cannot deprotonate 2,6-dimethoxyphenol.

(Choice C) Neither phosphatidylethanolamine nor 2,6-dimethoxyphenol will be extracted. Amines cannot be adequately protonated by weak acids such as H₂CO₃(aq), and NaHCO₃ is a weak base that will not deprotonate phenols.

Educational objective:

A mixture of organic compounds with acidic or basic functional groups can be separated by acid-base extraction. The basic or acidic molecules enter the aqueous layer as ionic salts by extraction with an acid or base, respectively. Amines are weak bases that require strong acids to be protonated. Phenols are weak acids that are only deprotonated by strong bases.

Time spent: 0 sec

Subject:
Organic Chemistry

QID: 401239

System:
Separation Techniques, Spectroscopy, and Analytical Methods

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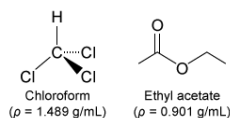


Figure 1 Structures and densities (ρ) of the B&D and GC organic solvents

Experiment 1

Y. lipolytica grown on rice straw containing 2,6-dimethoxyphenol were dried, then mixed with distilled water and 0.85% potassium chloride to create yeast extracts. Homogenized yeast extracts were transferred to separation funnels, and solutions with a 2:3.8 ratio of chloroform to methanol/water or ethyl acetate to ethanol/water were added. The resulting layers (L1, L2) in each extraction method were separated at the interface by films of cellular debris, as shown in Figure 2.

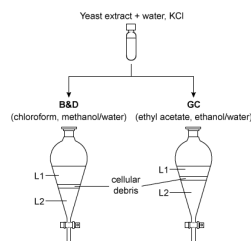


Figure 2 Separations obtained from B&D and GC lipid extraction methods

Experiment 2

Computational models indicate that glycerol (Figure 3) found in the yeast extract demonstrates partition differences between solvents in the B&D and GC methods.

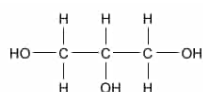


Figure 3 Structure of glycerol

To verify this prediction, chemists analyzed the B&D and GC solvent mixtures by proton nuclear magnetic resonance (^1H NMR) spectroscopy. A 160 mg/mL glycerol standard was separately prepared in each organic solvent from the B&D and GC methods. The glycerol standard was extracted with an equivalent volume of water/alcohol, then the organic layers were used to prepare NMR samples through iterative dilution. A 0.5 mL sample of organic layer was diluted with enough solvent to give a total volume of 10.0 mL, then 0.5 mL of this resulting solution was diluted with 0.5 mL of solvent to yield a final 1.0 mL NMR sample. The ^1H NMR spectra obtained from the two samples are shown in Figure 4 (hydroxyl hydrogens are not observed in the spectra).

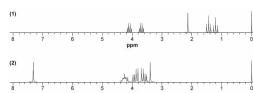
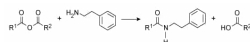


Figure 4 ^1H NMR spectra of the diluted organic solvent mixtures used in the (1) GC and (2) B&D extractions. The B&D mixture demonstrates the presence of glycerol.

The following reaction between an anhydride and phenylethylamine is conducted in organic solvent.

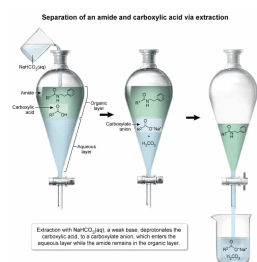


Which of the following aqueous solutions can be added to the reaction to separate the amide and carboxylic acid products through extraction? (Note: R^1 and R^2 = long hydrocarbon chains.)

- ☐ A. 5% NaCl [15%]
- ☐ B. 5% $\text{CH}_3\text{CO}_2\text{H}$ [20%]
- ☒ C. 5% NaHCO_3 [38%]
- ☐ D. 5% CH_3OH [25%]

Correct
39% answered correctly

Explanation:



Acylation reactions between **anhydrides** and amines generate **amides** and carboxylic acids. Amides and carboxylic acids with short hydrocarbon chains (R groups) are water-soluble molecules because they form hydrogen bonds in an aqueous solution. However, long hydrocarbon chains (R groups) on amides or carboxylic acids increase the hydrophobic nature of the molecules and make them insoluble in aqueous solvents.

Separation of amides and carboxylic acids can be accomplished by **extraction**. Carboxylic acids with long hydrocarbon chains can be induced to enter the aqueous layer if they are converted to an **ionic salt** via **deprotonation** by either a **strong base** or a **weak base**. The amide remains in the organic layer because it cannot be deprotonated by a weak base. Amide hydrolysis requires either a **strong acid** or **strong base** and heat.

The question shows that the reaction between the anhydride and phenylethylamine produces an amide and a carboxylic acid. To separate the products of this reaction, extraction with a **weak base** such as 5% NaHCO_3 will **deprotonate the carboxylic acid**. The resulting carboxylate anion will enter the polar aqueous layer, and the amide will remain in the organic layer.

(Choice A) 5% NaCl is a neutral salt solution and does not react with amides or carboxylic acids. Saturated NaCl is sometimes used in extractions to prevent emulsions or remove residual water droplets from the organic layer.

(Choice B) 5% $\text{CH}_3\text{CO}_2\text{H}$ is a **weak acid**. Weak acids do not react with carboxylic acids, and weak acids cannot significantly protonate amides. Therefore, 5% $\text{CH}_3\text{CO}_2\text{H}$ cannot be used to separate amides and carboxylic acids.

(Choice D) CH_3OH is a polar organic solvent that is miscible with water. A 5% solution of CH_3OH in water does not react with either carboxylic acids or amides.

Educational objective:

Separation of amides and carboxylic acids can be done via extraction. Treatment with a strong or weak base will convert carboxylic acids to a carboxylate salt, which dissolves in the aqueous layer during extraction. The amide will remain in the organic layer.

Time spent: 0 sec
Subject:

Organic Chemistry

QID: 401240

System:

Separation Techniques, Spectroscopy, and Analytical Methods

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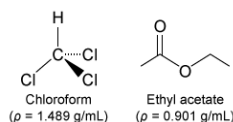


Figure 1 Structures and densities (ρ) of the B&D and GC organic solvents

Experiment 1

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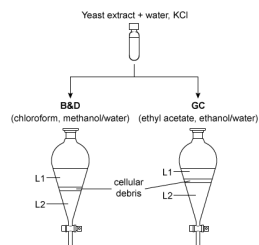


Figure 2 Separations obtained from B&D and GC lipid extraction methods

Experiment 2

Computational models indicate that glycerol (Figure 3) found in the yeast extract demonstrates partition differences between solvents in the B&D and GC methods.

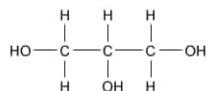


Figure 3 Structure of glycerol

To verify this prediction, chemists analyzed the B&D and GC solvent mixtures by proton nuclear magnetic resonance (^1H NMR) spectroscopy. A 160 mg/mL glycerol standard was separately prepared in each organic solvent from the B&D and GC methods. The glycerol standard was extracted with an equivalent volume of water/alcohol, then the organic layers were used to prepare NMR samples through iterative dilution. A 0.5 mL sample of organic layer was diluted with enough solvent to give a total volume of 10.0 mL, then 0.5 mL of this resulting solution was diluted with 0.5 mL of solvent to yield a final 1.0 mL NMR sample. The ^1H NMR spectra obtained from the two samples are shown in Figure 4 (hydroxyl hydrogens are not observed in the spectra).

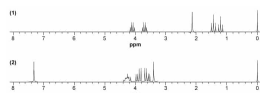


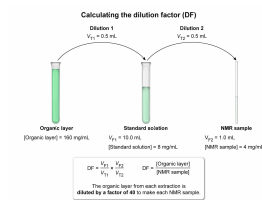
Figure 4 ^1H NMR spectra of the diluted organic solvent mixtures used in the (1) GC and (2) B&D extractions. The B&D mixture demonstrates the presence of glycerol.

Based on the dilutions described in Experiment 2, the samples obtained from the organic solvent layer were diluted by what factor to prepare the NMR samples?

- ☐ A. 2 [5%]
☐ B. 20 [34%]
☒ C. 40 [55%]
☐ D. 80 [3%]

Correct
 55% answered correctly

Explanation:



For the preparation of the NMR samples described in Experiment 2, two methods can be used to calculate the magnitude by which the organic solvent mixture from each extraction was diluted.

Method 1

The **dilution factor** (DF) can be calculated for each dilution step by **dividing the final volume** of the dilute solution V_F by the **volume** of solute solution **transferred** V_T .

$$DF = \frac{V_F}{V_T}$$

The passage states that 0.5 mL (V_{T1}) of the organic solvent mixture was diluted in 9.5 mL of solvent to produce a standard solution ($V_{F1} = 10.0$ mL). Then, 0.5 mL (V_{T2}) of the resulting standard solution was taken and diluted in 0.5 mL of solvent to give the NMR sample ($V_{F2} = 1.0$ mL). Multiplying the DF of both steps together yields the overall DF.

$$DF = \frac{V_{F1}}{V_{T1}} \times \frac{V_{F2}}{V_{T2}}$$

$$DF = \frac{10.0 \text{ mL}}{0.5 \text{ mL}} \times \frac{1.0 \text{ mL}}{0.5 \text{ mL}}$$

$$DF = 20 \times 2 = 40$$

Method 2

The DF can be alternatively calculated as a **ratio of the initial concentration to the final concentration**. First, the concentration of the standard solution is calculated by determining the mass of glycerol transferred from the organic solvent mixture to the standard solution and then dividing this mass by the final volume of the standard solution (10.0 mL).

$$\text{Mass transferred (first dilution)} = [\text{Organic solvent mixture}] \times \text{Volume transferred}$$

$$\text{Mass transferred (first dilution)} = \frac{160 \text{ mg}}{\text{mL}} \times 0.5 \text{ mL transferred} = 80 \text{ mg}$$

$$[\text{Standard solution}] = \frac{\text{Mass transferred}}{\text{Volume of standard solution}} = \frac{80 \text{ mg}}{10 \text{ mL}} = 8 \frac{\text{mg}}{\text{mL}}$$

Next, the concentration of the NMR sample is calculated in a similar fashion by determining the mass of glycerol transferred from the standard solution and then dividing this mass by the final volume of the NMR sample (1.0 mL).

$$\text{Mass transferred (second dilution)} = [\text{Standard solution}] \times \text{Volume transferred}$$

$$\text{Mass transferred (second dilution)} = \frac{8 \text{ mg}}{\text{mL}} \times 0.5 \text{ mL transferred} = 4 \text{ mg}$$

$$[\text{NMR sample}] = \frac{\text{Mass transferred}}{\text{Volume of NMR solution}} = \frac{4 \text{ mg}}{1.0 \text{ mL}} = 4 \frac{\text{mg}}{\text{mL}}$$

Finally, the DF is calculated.

$$\text{DF} = \frac{[\text{organic layer}]}{[\text{NMR sample}]} = \frac{160 \text{ mg/mL}}{4 \text{ mg/mL}} = 40$$

Therefore, the samples from the organic solvent mixture were diluted by a factor of 40 to make the NMR samples.

(Choice A) This value is the DF of the second step (ie, $\text{DF} = \frac{V_{F2}}{V_{T2}} = \frac{1.0 \text{ mL}}{0.5 \text{ mL}} = 2$).

(Choice B) This value is the DF of the first step (ie, $\text{DF} = \frac{V_{F1}}{V_{T1}} = \frac{10 \text{ mL}}{0.5 \text{ mL}} = 20$).

(Choice D) This value is obtained by incorrectly using a V_F of 2.0 mL for the second step and then multiplying by the correct DF of the first step (20).

Educational objective:

The dilution factor (DF) of each step in a serial dilution is calculated by dividing the final volume of the dilute solution V_F by the volume of the solute solution transferred V_T . Multiplying together the dilution factors from each dilution step yields the overall DF of a dilution sequence.

Time spent: 0 sec
Subject:
Organic Chemistry

QID:401241
System:
Separation Techniques, Spectroscopy, and Analytical Methods

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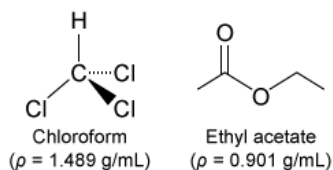


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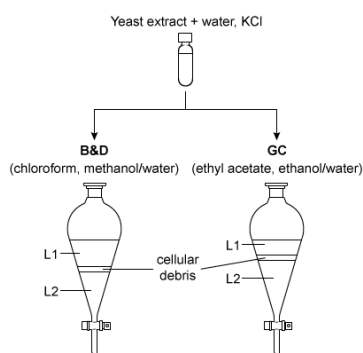


Figure 2 Separations obtained from B&D and GC lipid extraction methods

Experiment 2

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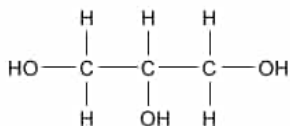


Figure 3 Structure of glycerol

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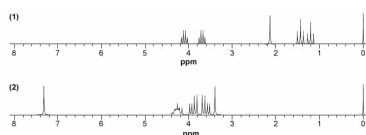


Figure 4 ^1H NMR spectra of the diluted organic solvent mixtures used in the (1) GC and (2) B&D extractions. The B&D mixture demonstrates the presence of glycerol.

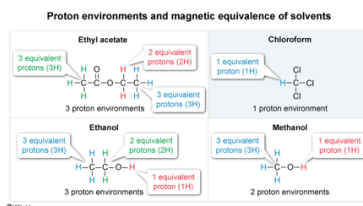
Signals with a relative integration greater than 1 are present in the ^1H NMR spectrum for each of the following compounds EXCEPT:

- ☐ A. Ethyl acetate [9%]

- ✓ ☒ B. Chloroform [68%]
☐ C. Ethanol [6%]
☐ D. Methanol [15%]

Correct
 68% answered correctly

Explanation:



When hydrogen atoms enter an external magnetic field (B_0), their nuclear spin either aligns with or against the direction of B_0 . Nuclei aligned with the magnetic field are in a low-energy alpha spin state whereas those aligned against the field are in a high-energy beta spin state. Proton **nuclear magnetic resonance spectroscopy** (^1H NMR) detects hydrogen atoms by exposing them to **radio waves** that excite atoms from the alpha spin to the beta spin state.

The difference in energy between the alpha and beta spin states is the **resonance energy** and is proportional to the **effective magnetic field** B_{eff} experienced by the hydrogen nucleus. Electrons generate magnetic fields B_{ind} that **shield** a nucleus from the external magnetic field. Consequently, the B_{eff} of each hydrogen atom depends on the **electronic environment** that surrounds it.

$$B_{\text{eff}} = B_0 - B_{\text{ind}}$$

The position of a hydrogen peak in the ^1H NMR spectrum is referred to as the chemical shift, which is proportional to B_{eff} .

Hydrogen atoms on a molecule that display **rotational or planar symmetry** and are in identical magnetic environments will have identical chemical shifts. The relative **integration** of a ^1H NMR signal is proportional to the number of protons in that magnetic environment.

To predict the number of unique signals on the ^1H NMR spectrum, the number of magnetic environments must be determined. Figure 1 shows that chloroform has one hydrogen and three chlorine atoms attached to a central carbon. As there is only one proton in chloroform, it will inherently have a single peak in its ^1H NMR spectrum. Therefore, chloroform is the only compound that does not contain a relative integration greater than 1.

(Choice A) Ethyl acetate has three unique proton environments, which each contain between two and three equivalent protons.

(Choice C) Ethanol has three unique proton environments, which each contain between one and three equivalent protons.

(Choice D) Methanol has two unique proton environments, which contain either one or three equivalent protons.

Educational objective:

Hydrogen atoms on a molecule that display rotational or planar symmetry are in identical magnetic environments. Equivalent hydrogens have identical chemical shifts and are represented as a single signal on the ^1H NMR spectrum. The relative integration of a ^1H NMR signal is proportional to the number of protons in a magnetic environment.

Time spent: 0 sec
 Subject:
 Organic Chemistry

QID: 401242
 System:
 Separation Techniques, Spectroscopy, and Analytical Methods

Lipid extractions are typically done by the Bligh and Dyer (B&D) method, which uses a water/methanol mixture and chloroform for the two phases of the solvent partition. Researchers specializing in green chemistry (GC) have proposed that greener solvents such as ethyl acetate and ethanol can be used to replace the toxic chloroform and methanol, respectively, from the B&D method. The following experiments test this proposal by comparing the B&D and GC solvent methods (Figure 1) to separate lipids from an extract of *Yarrowia lipolytica* yeast cells.

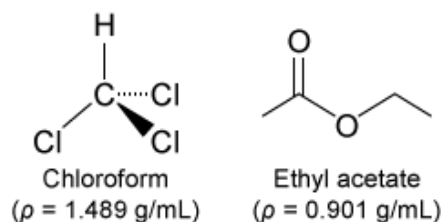


Figure 1 Structures and densities (ρ) of the B&D and GC organic solvents

Experiment 1

Y. lipolytica grown on rice straw containing 2,6-dimethoxyphenol were dried, then mixed with distilled water and 0.85% potassium chloride to create yeast extracts. Homogenized yeast extracts were transferred to separation funnels, and solutions with a 2:3.8 ratio of chloroform to methanol/water or ethyl acetate to ethanol/water were added. The resulting layers (L1, L2) in each extraction method were separated at the interface by films of cellular debris, as shown in Figure 2.

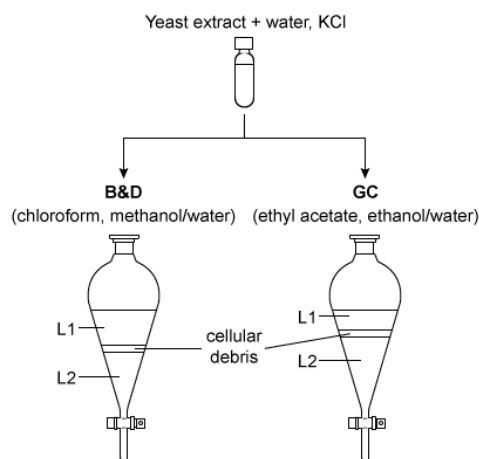


Figure 2 Separations obtained from B&D and GC lipid extraction methods

Experiment 2

Computational models indicate that glycerol (Figure 3) found in the yeast extract demonstrates partition differences between solvents in the B&D and GC methods.

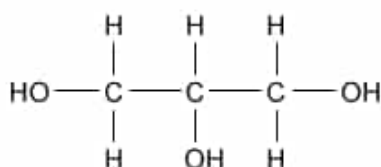


Figure 3 Structure of glycerol

To verify this prediction, chemists analyzed the B&D and GC solvent mixtures by proton nuclear magnetic

resonance (^1H NMR) spectroscopy. A 160 mg/mL glycerol standard was separately prepared in each organic solvent from the B&D and GC methods. The glycerol standard was extracted with an equivalent volume of water/alcohol, then the organic layers were used to prepare NMR samples through iterative dilution. A 0.5 mL sample of organic layer was diluted with enough solvent to give a total volume of 10.0 mL, then 0.5 mL of this resulting solution was diluted with 0.5 mL of solvent to yield a final 1.0 mL NMR sample. The ^1H NMR spectra obtained from the two samples are shown in Figure 4 (hydroxyl hydrogens are not observed in the spectra).

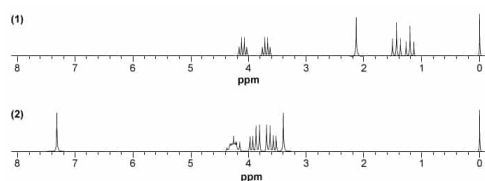


Figure 4 ^1H NMR spectra of the diluted organic solvent mixtures used in the (1) GC and (2) B&D extractions. The B&D mixture demonstrates the presence of glycerol.

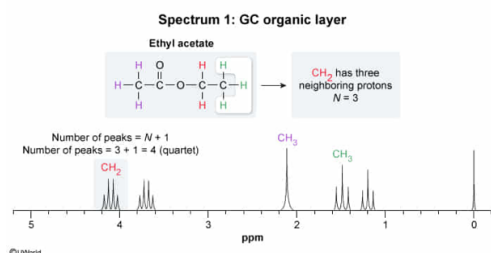
Which of the following ^1H NMR signals in Figure 4 most likely corresponds to a proton environment in ethyl acetate?

- ☐ A. The singlet at 0 ppm in the spectrum of the organic layer from the GC extraction [4%]
- ☒ B. The quartet at 4.1 ppm in the spectrum of the organic layer from the GC extraction [71%]
- ☐ C. The singlet at 7.3 ppm in the spectrum of the organic layer from the B&D extraction [14%]
- ☐ D. The multiplet at 4.3 ppm in the spectrum of the organic layer from the B&D extraction [10%]

Correct

70% answered correctly

Explanation:



Proton **nuclear magnetic resonance spectroscopy** (^1H NMR) detects hydrogen atoms (protons) in a molecule by applying an **external magnetic field** B_0 and radio frequencies to a sample. Protons that exist in different magnetic environments are referred to as magnetically nonequivalent hydrogens. Interactions between neighboring, nonequivalent protons result in the coupling of the protons, causing **spin-spin splitting** of peaks in the ^1H NMR spectrum. The **splitting pattern** of a given proton can generally be determined by the **$N + 1$ rule**, where N is the number of nonequivalent protons on adjacent carbon atoms.

Ethyl acetate has three sets of nonequivalent protons:

- One $-\text{CH}_3$ group bonded to a carbonyl ($\text{C}=\text{O}$) ($N + 1 = 1$, **singlet**)
- One $-\text{CH}_3$ group bonded to a $-\text{CH}_2-$ group ($N + 1 = 3$, **triplet**)
- One $-\text{CH}_2-$ group bonded to an oxygen atom and a $-\text{CH}_3$ group (three neighboring protons, $N + 1 = 4$, **quartet**)

The **chemical shift** of the signals depends on the electronic environment of the protons. The two -CH_3 groups are surrounded by an electron cloud and shielded from B_0 , resulting in upfield signals (ie, smaller chemical shift). The triplet appears at 1.3 ppm, and the singlet appears farther downfield (ie, larger chemical shift) at 2.1 ppm because this -CH_3 group is deshielded by the adjacent carbonyl. The quartet appears farther downfield than the -CH_3 groups at 4.1 ppm because the $\text{-CH}_2\text{-}$ group is deshielded by the adjacent electronegative oxygen atom.

Therefore, the **quartet at 4.1 ppm** in Spectrum 1 is the only available choice with a signal corresponding to ethyl acetate.

(Choice A) The **singlet at 0 ppm** in Spectrum 1 is from the four -CH_3 groups on tetramethylsilane (TMS), an internal standard used to calibrate the chemical shifts of an NMR sample.

(Choices C and D) These signals cannot correspond to ethyl acetate because ethyl acetate was not used in the B&D extraction. In Spectrum 2, the **multiplet** at 4.3 ppm corresponds to the central H atom in glycerol, and the singlet at 7.3 ppm corresponds to the hydrogen in chloroform, which is **deshielded** due to neighboring electron-withdrawing chlorine atoms.

Educational objective:

Spin-spin splitting of peaks in an NMR spectrum results from interactions between neighboring nonequivalent protons. The splitting pattern can be determined by the $N + 1$ rule, where N is the number of neighboring nonequivalent protons.

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